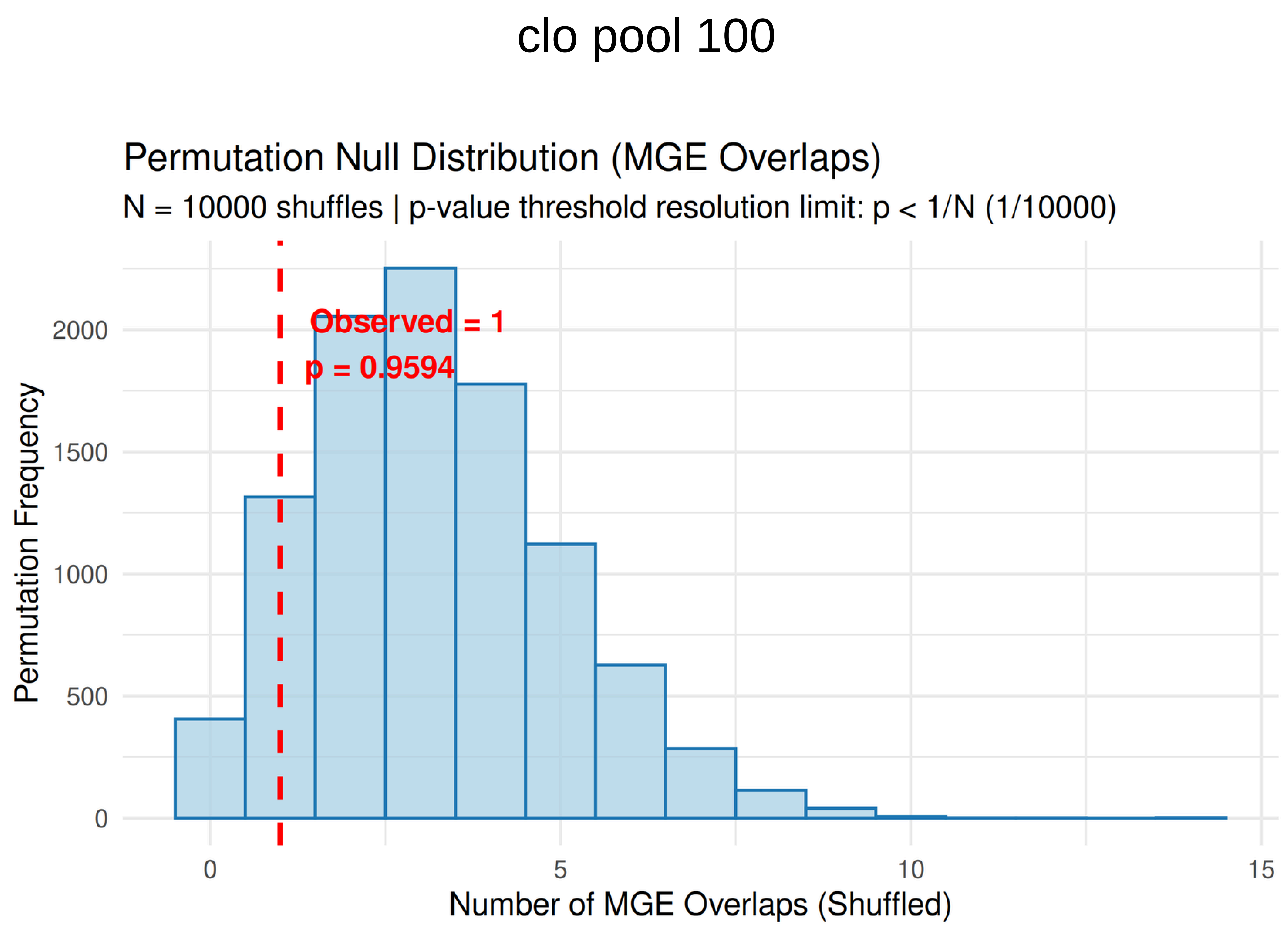
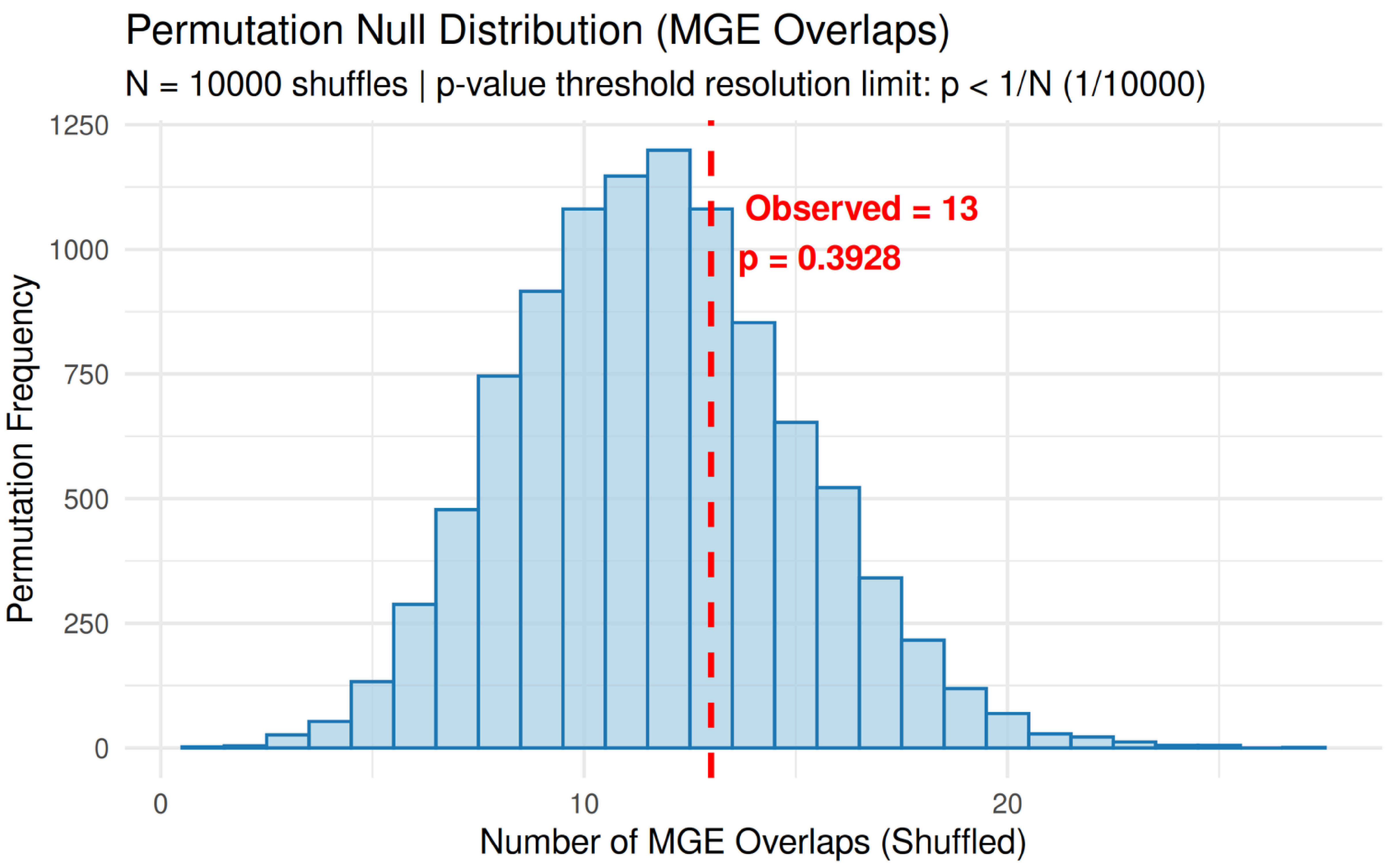
mobile genetic element permutation test by using gin for C. difficile, M. tuberculosis and S. enterica subsp. enterica

dataset	pool	observed	permutations	ge_observed	p value
clo	100	1	10000	9594	0.96
clo	1000	0	0	0	1
mtb	100	13	10000	3927	0.39
mtb	1000	0	0	0	1
sen	100	0	10000	10000	1
sen	1000	0	10000	10000	1


clo pool 100
Permutation Null Distribution (MGE Overlaps)
N = 10000 shuffles | p-value threshold resolution limit: p < 1/N (1/10000)
Observed = 1
p = 0.9594
Permutation Frequency
2000
1500
1000
500
0
0
5
10
15
Number of MGE Overlaps (Shuffled)

mtb pool 100



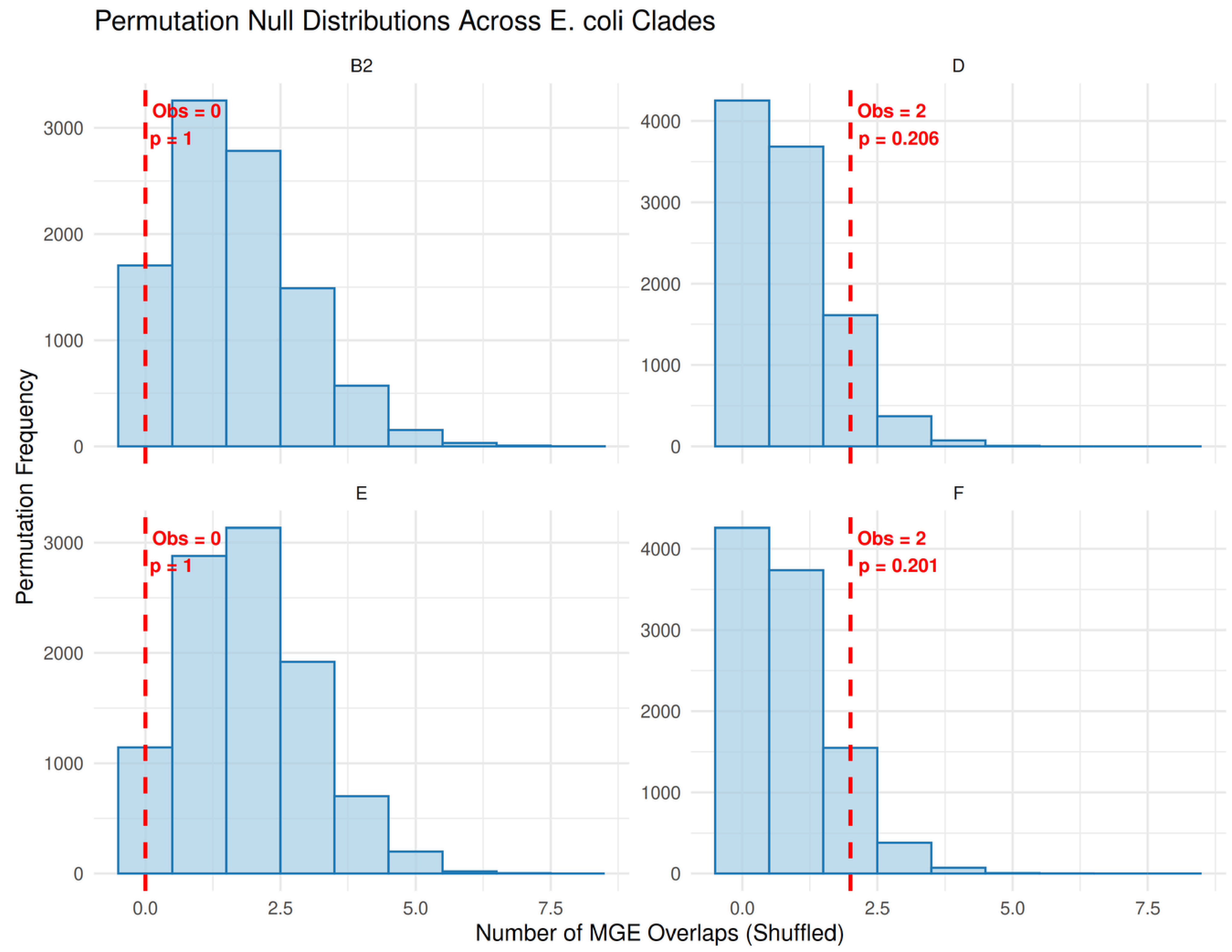
Permutation Null Distribution (MGE Overlaps)
N = 10000 shuffles | p-value threshold resolution limit: p < 1/N (1/10000)
Observed = 13
p = 0.3928
Permutation Frequency
1250
1000
750
500
250
0
0
10
20
Number of MGE Overlaps (Shuffled)

mobile genetic element permutation test by using gin for 6 strains of E. coli

for fur only

dataset	observed	permutations	ge_observed	p value
b2	0	10000	10000	1
d	2	10000	2060	0.2
e	0	10000	10000	1
f	2	10000	2007	0.2

a and b1 strains did not have any markers

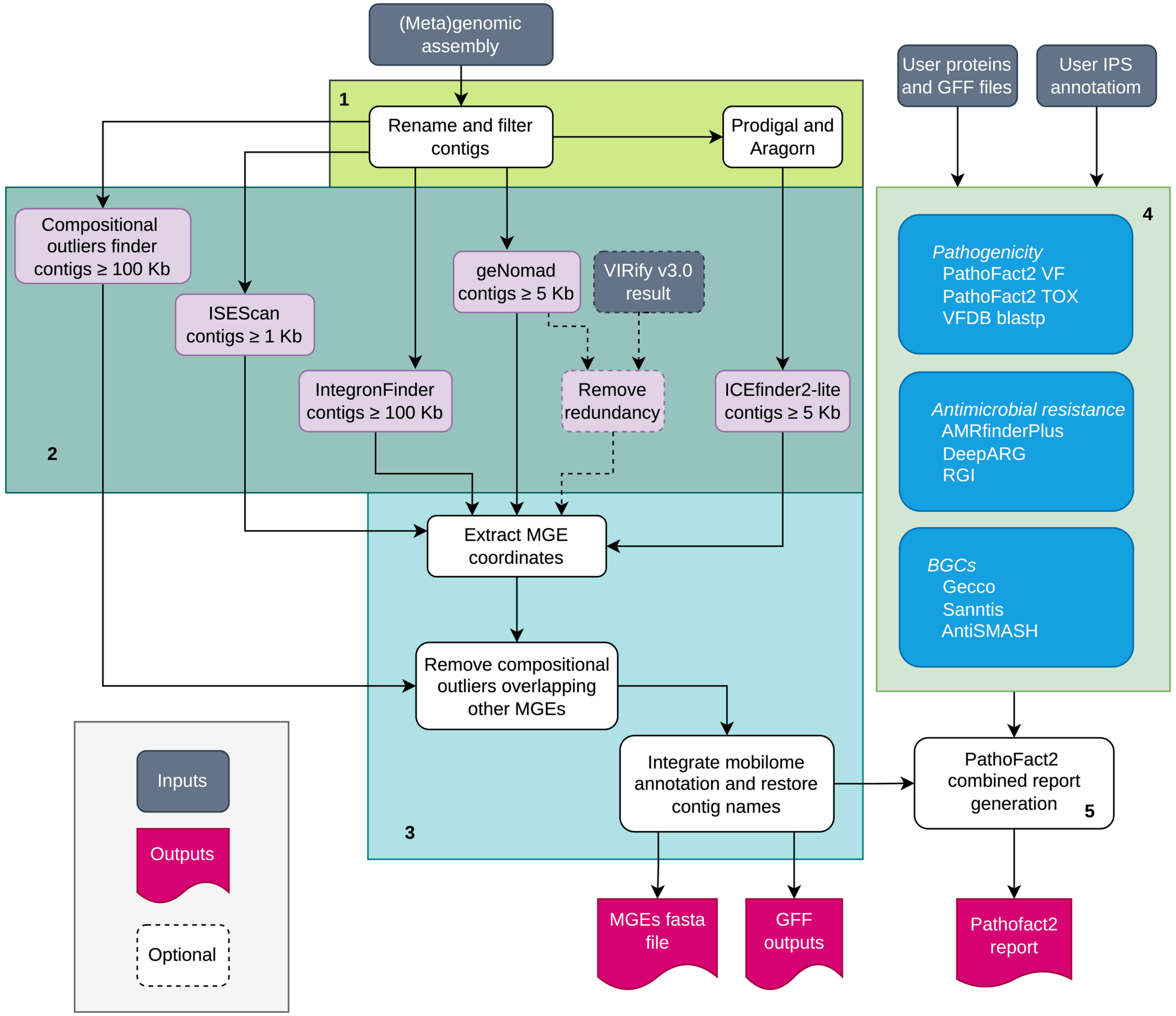


Permutation Null Distributions Across E. coli Clades
B2
D
E
F
Obs = 0
p = 1
Obs = 2
p = 0.206
Obs = 0
p = 1
Obs = 2
p = 0.201
Permutation Frequency
Number of MGE Overlaps (Shuffled)
0
1000
2000
3000
4000
0.0
2.5
5.0
7.5

mobile genetic element permutation test by using mobilome annotation pipeline
for 6 strains of E. coli

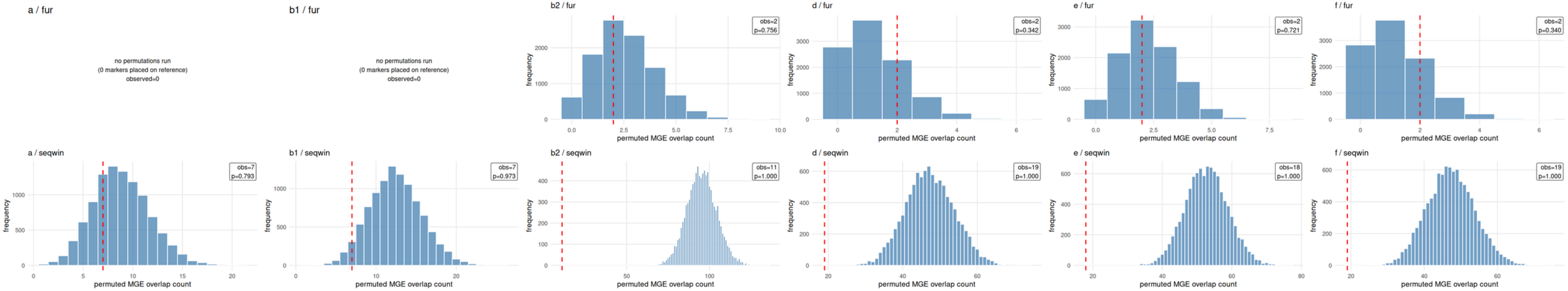
this time i did it for both fur and seqwin

mobilome annotation pipeline

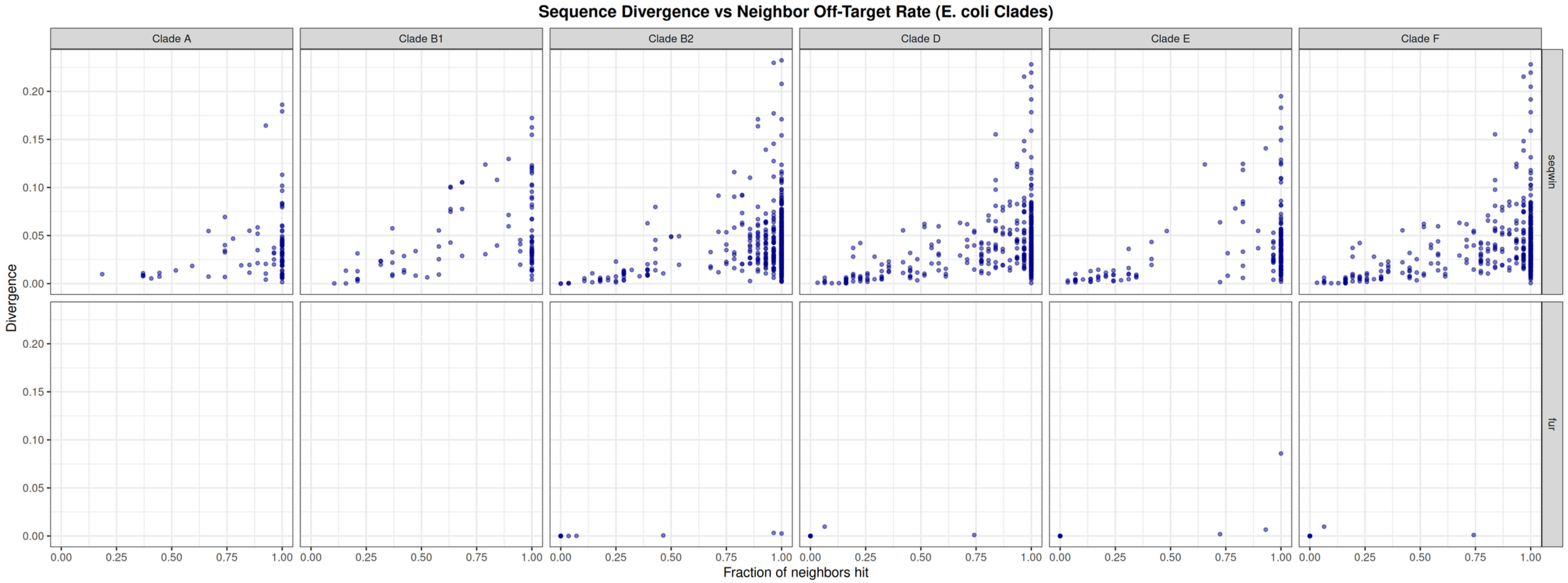


strain	marker_type	observed	permutations	observed	p-value
a	fur	0	0	0	1
a	seqwin	7	10000	7930	0.79
b1	fur	0	0	0	1
b1	seqwin	7	10000	9734	0.97
b2	fur	2	10000	7561	0.75
b2	seqwin	11	10000	10000	1
d	fur	2	10000	3417	0.34
d	seqwin	19	10000	10000	1
e	fur	2	10000	7207	0.72
e	seqwin	18	10000	10000	1
f	fur	2	10000	3400	0.34
f	seqwin	19	10000	10000	1

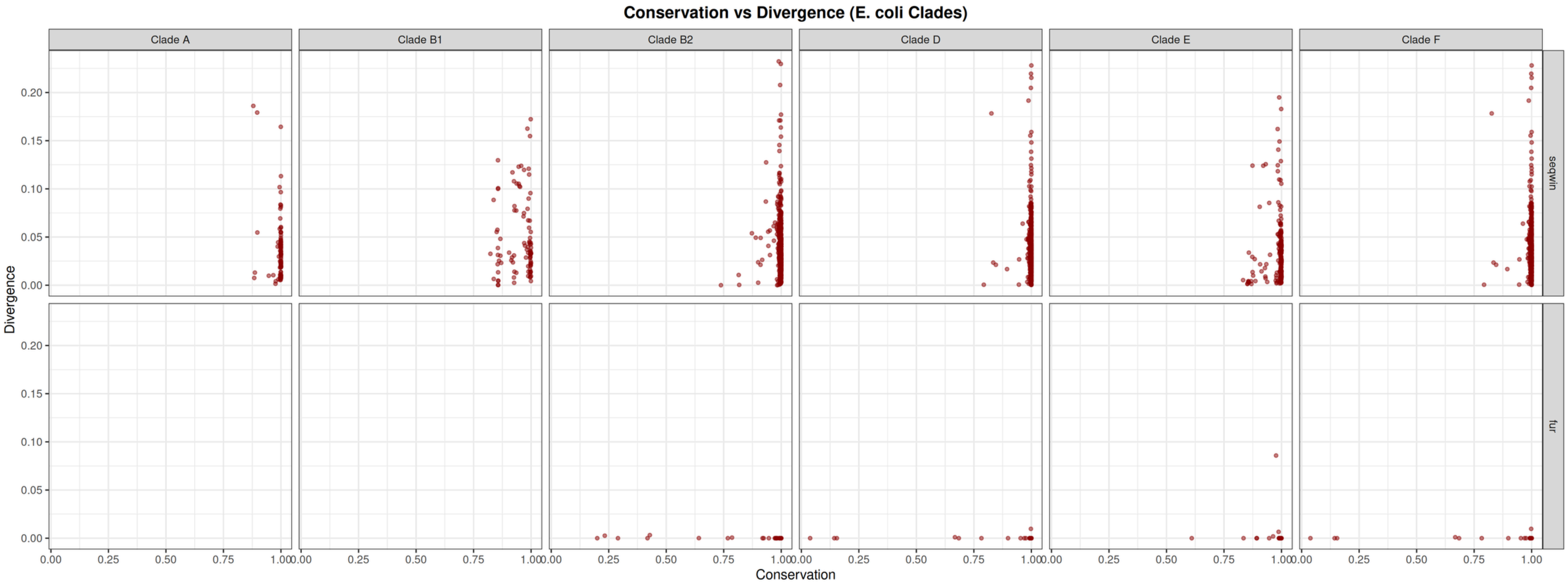
Null distributions: permuted MGE overlaps vs. observed (fur vs. seqwin)



replicating seqwin paper’s supplementary information tables for 6 strains of E. coli



Sequence Divergence vs Neighbor Off-Target Rate (E. coli Clades)
Clade A
Clade B1
Clade B2
Clade D
Clade E
Clade F
seqwin
fur
Divergence
0.00
0.05
0.10
0.15
0.20
Fraction of neighbors hit
0.00
0.25
0.50
0.75
1.00



Conservation vs Divergence (E. coli Clades)
Clade A
Clade B1
Clade B2
Clade D
Clade E
Clade F
seqwin
fur
Divergence
0.00
0.05
0.10
0.15
0.20
Conservation
0.00
0.25
0.50
0.75
1.00